MODULE 4 — IMAGE SEGMENTATION

1. Introduction to Image Segmentation

Definition

Image segmentation is the process of dividing an image into meaningful regions or groups of pixels based on properties such as intensity, colour, texture, or boundaries.

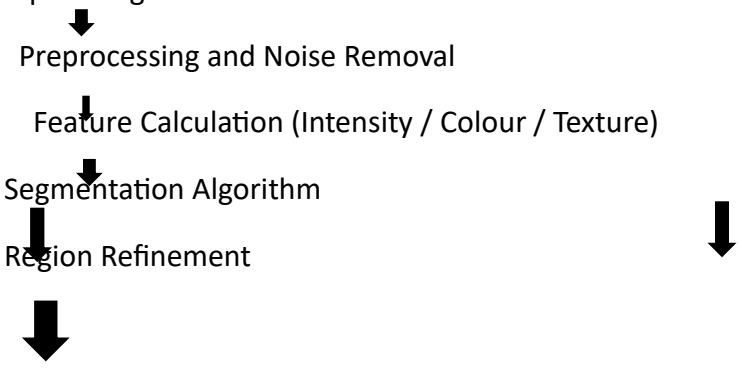
The objective is to simplify an image and make it easier to identify individual objects or regions.

For example, in a medical image, segmentation can separate a tumour from surrounding tissue. In autonomous driving, it can distinguish roads, vehicles, pedestrians, and buildings.

Types of image segmentation

- Threshold-based: separates pixels according to intensity values.
- Edge-based: uses detected boundaries to identify regions.
- Region-based: groups neighbouring pixels with similar properties.
- Clustering-based: groups pixels according to feature similarity.
- Graph-based: models pixels or regions as graph nodes.
- Texture-based: separates regions using texture descriptors.
- Object segmentation: identifies the pixels belonging to specific objects.

General block diagram



Advantages

- Separates objects from backgrounds.
- Supports object recognition and measurement.
- Helps identify regions of interest.

- Reduces the complexity of subsequent analysis.

Disadvantages

- Noise can produce incorrect regions.
- Shadows and illumination changes may affect boundaries.
- Similar objects may merge into one region.
- Complex images can require substantial computation.

Applications

Medical image analysis, satellite imagery, autonomous vehicles, face analysis, industrial inspection, and agricultural monitoring.

2. Region Growing

Definition

Region growing is a region-based segmentation technique that starts with one or more seed pixels and expands each region by adding neighbouring pixels that satisfy a similarity criterion.

The similarity may be based on intensity, colour, texture, or a statistical measure.

Principle

A pixel is added to a region if it is sufficiently similar to the region's seed or current region statistics.

For example, a seed placed inside a white object may expand through neighbouring white pixels until it reaches the object's boundary.

Block diagram

Input Image

↓

Select Seed Pixel(s)

↓

Examine Neighbouring Pixels

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Test Similarity Criterion

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Add Similar Pixels to Region

↓

Update Region and Neighbours

↓

Stop When No More Pixels Qualify

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Segmented Regions

Algorithm

Input: Image I, seed pixels S, threshold T

1. Initialize an empty region R.

2. Add seed pixel(s) to R.

3. Place neighbouring pixels in a candidate queue.

4. While the queue is not empty:

    Remove a candidate pixel p.

    If p satisfies the similarity criterion:

        Add p to R.

        Add its unvisited neighbours to the queue.

5. Mark all accepted pixels as the segmented region.

Output: Region labels

A simple intensity criterion is:

$$ |I(p)-\mu_R|\leq T $$

where $I(p)$ is the candidate pixel intensity and μ_R is the current region's mean intensity.

Advantages

- Simple to understand and implement.
- Produces connected regions.
- Can use multiple seeds for multiple objects.
- Works well when the target region is relatively uniform.

Disadvantages

- Seed selection affects the result.
- Noise can cause leakage into neighbouring regions.
- Threshold selection is important.
- Large images may require substantial computation.

Applications

Tumour segmentation, organ extraction, agricultural images, defect detection, and object-region extraction.

3. Edge-Based Image Segmentation

Definition

Edge-based segmentation identifies boundaries between regions by detecting strong changes in intensity, colour, or texture. The detected edges are then used to separate objects from their backgrounds or from one another.

Working

1. Convert the image to an appropriate colour or grayscale representation.
2. Reduce noise using smoothing.
3. Apply an edge detector such as Canny, Sobel, or LoG.
4. Connect broken edge fragments where possible.
5. Close gaps and construct boundaries.
6. Fill closed boundaries or label the enclosed regions.

Block diagram

Input Image

Noise Reduction

Edge Detection

Edge Linking and Gap Closing

Boundary Formation

Region Labelling / Filling

Algorithms

Common edge detectors used for segmentation include:

- Sobel: estimates first-order gradients.
- Canny: combines smoothing, gradient detection, non-maximum suppression, and hysteresis.
- LoG: identifies zero crossings after Gaussian smoothing.

Morphological operations such as dilation and closing may be used to connect boundary gaps.

Advantages

- Effective when objects have strong boundaries.

- Provides accurate outlines in many high-contrast images.
- Useful for shape extraction and measurement.

Disadvantages

- Weak boundaries may be missed.
- Noise can generate false edges.
- Broken boundaries make region filling difficult.
- Textured regions may contain many unwanted edges.

Applications

Industrial object inspection, road boundaries, medical structures, document analysis, and shape measurement.

4. Graph-Cut Segmentation

Definition

Graph-cut segmentation models an image as a graph. Pixels or groups of pixels are represented by nodes, and edges connect neighbouring nodes. The algorithm assigns labels by minimizing an energy function.

For example, pixels can be labelled as foreground or background.

Principle

The energy commonly combines two terms:

$$ E(L)= \sum_p D_p(L_p) +\lambda\sum_{(p,q)}V_{pq}(L_p,L_q) $$

where:

- L_p is the label assigned to pixel p.
- D_p is the data cost for assigning that label.
- V_{pq} is a pairwise cost encouraging compatible neighbouring labels.
- λ controls the relative importance of the smoothness term.

The minimum-energy labelling can be obtained using a graph-cut algorithm for suitable energy formulations.

Block diagram

Input Image

Define Foreground and Background Information

Construct Graph and Energy Function

Calculate Data and Boundary Costs

Find Minimum Cut

Assign Pixel Labels

Segmented Foreground / Background

Working

1. Define foreground and background labels, often using user-provided seeds.
2. Construct a graph representing pixel or region relationships.
3. Assign costs based on image evidence and label consistency.
4. Calculate a cut separating foreground and background.
5. Convert the minimum-cut result into a segmented image.
6. Refine the result if necessary.

Advantages

- Combines pixel appearance and neighbourhood relationships.
- Can produce accurate foreground extraction.
- Incorporates prior knowledge through seeds.
- Handles many difficult boundaries better than simple thresholding.

Disadvantages

- Can require substantial memory for large images.
- Results depend on energy terms and parameter settings.
- Some implementations require user input.
- General energy minimization may not be solvable exactly by a standard minimum cut.

Applications

Image editing, background removal, medical image segmentation, object extraction, and interactive photo selection.

5. Mean-Shift Segmentation

Definition

Mean-shift segmentation is a non-parametric clustering method that groups pixels by locating dense regions, or modes, in a feature distribution. Features can include colour, spatial position, and texture.

It does not require the number of clusters to be specified in advance.

Principle

Each pixel is represented in a feature space, often using position and colour:

\[z_p=(x_p,y_p,R_p,G_p,B_p) \]

The mean-shift vector moves a feature point towards a region of higher data density. Points converging to the same mode are grouped together.

Block diagram

Input Image

Represent Pixels by Position and Colour

Choose Spatial and Colour Bandwidths

Compute Mean-Shift Vectors

Iteratively Move Points Toward Density Modes

Merge Similar Modes

Assign Segment Labels

Algorithm

1. Represent every pixel by its selected features.
2. Select bandwidths for spatial and feature similarity.
3. Calculate the local weighted mean around each point.
4. Shift the point towards that mean.
5. Repeat until convergence.
6. Merge points that converge to similar modes.
7. Label pixels according to their converged modes.

Advantages

- Does not require a predefined cluster count.
- Can identify irregularly shaped clusters in feature space.
- Combines spatial and colour information.

- Useful for colour image segmentation.

Disadvantages

- Computationally expensive for large images.
- Sensitive to bandwidth selection.
- May merge different regions or split uniform regions.
- Processing time can be high for dense feature spaces.

Applications

Colour image segmentation, object tracking, image smoothing, video analysis, and region grouping.

6. MRF — Markov Random Field Segmentation

Definition

A Markov Random Field (MRF) is a probabilistic model used to represent dependencies among neighbouring pixels or regions. In image segmentation, it assigns labels while considering both the observed image data and spatial relationships between pixels.

Principle

The label of a pixel depends on its observed features and the labels of neighbouring pixels.

The objective is often expressed as an energy minimization problem:

$$ E(L)=\sum_p D_p(L_p) +\lambda\sum_{(p,q)}V_{pq}(L_p,L_q) $$

Here, D_p represents how well the label matches the pixel's observed data, while V_{pq} encourages spatial consistency.

Block diagram

Input Image

Extract Pixel Features

Define Label Set and Neighbourhood

Construct MRF / Energy Model

Optimize Label Assignment

Obtain Segmented Image

Working

1. Extract intensity, colour, or texture features.

2. Define possible labels for each pixel.
3. Calculate data likelihoods or data costs.
4. Define neighbourhood relationships and smoothness costs.
5. Optimize the joint labelling using an appropriate method, such as graph cuts, belief propagation, or iterative conditional modes.
6. Assign the final label to each pixel.

Advantages

- Incorporates spatial context.
- Reduces isolated, inconsistent labels.
- Can combine multiple types of image evidence.
- Useful for noisy or ambiguous images.

Disadvantages

- Model design can be complicated.
- Inference can be computationally expensive.
- Incorrect assumptions about neighbouring pixels may oversmooth boundaries.
- Results depend on the data and smoothness terms.

Applications

Medical image segmentation, satellite image classification, denoising, texture segmentation, and scene labelling.

Important distinction: MRF is a probabilistic modelling framework. Graph cuts are an optimization technique that can solve certain MRF energy formulations; they are related but not identical concepts.

7. Texture Segmentation

Definition

Texture segmentation divides an image into regions according to differences in surface patterns, such as roughness, regularity, orientation, or spatial frequency.

It is particularly useful when two regions have similar average colours but different textures.

Texture features

Common texture features include:

- Local intensity statistics: mean and variance.
- Gray-Level Co-occurrence Matrix (GLCM) features: contrast, homogeneity, energy, and correlation.
- Gabor filter responses.
- Local Binary Patterns (LBP).
- Wavelet coefficients from DWT.

Block diagram

Input Image

Divide into Local Windows or Regions

Extract Texture Features

Compare Feature Vectors

Cluster or Classify Regions

Texture-Segmented Image

Working

1. Divide the image into local neighbourhoods or pixels.
2. Extract texture descriptors.
3. Construct a feature vector for each region.
4. Compare or cluster the feature vectors.
5. Assign labels to similar texture regions.
6. Refine boundaries using spatial information.

Advantages

- Separates regions with similar colours but different patterns.
- Useful for natural and industrial images.
- Supports material and surface classification.

Disadvantages

- Requires suitable feature selection.
- Window size influences spatial accuracy.
- Texture changes within one object may cause over-segmentation.

- Computational cost depends on the descriptor.

Applications

Fabric inspection, wood-grain analysis, agricultural monitoring, medical images, and remote sensing.

8. Object Detection

Definition

Object detection is a computer vision task that identifies objects in an image and estimates their locations, usually using bounding boxes and class labels.

Difference from segmentation: object detection generally identifies object locations, while segmentation assigns labels at the pixel or region level.

Types

- Traditional detection: sliding windows, HOG descriptors, and classifiers such as SVM.
- Two-stage detection: methods such as Faster R-CNN generate candidate regions and classify them.
- One-stage detection: methods such as YOLO predict objects directly in a single main detection pass.

General block diagram

Input Image

Preprocessing / Feature Extraction

Object Detection Model

Class and Bounding-Box Prediction

Confidence Filtering and NMS

Detected Objects

Non-Maximum Suppression (NMS)

When several bounding boxes overlap around the same object, NMS keeps the higher-confidence box and removes redundant boxes with excessive overlap.

Intersection over Union (IoU) is:

\[IoU=\frac{\text{Area of Intersection}}{\text{Area of Union}} \]

Advantages

- Detects and locates multiple objects.

- Enables automated scene understanding.
- Supports real-time systems with suitable models and hardware.

Disadvantages

- Requires training data for learned models.
- Small, occluded, or overlapping objects can be difficult to detect.
- Accuracy depends on lighting, scale, and training distribution.
- Deep learning models can require substantial computation.

Applications

Surveillance, autonomous driving, industrial inspection, healthcare, retail analytics, and robotics.